

SentinelPulse-LAR: Identity-Preserving Activation with Learnable Localized Corrections

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Abstract—Traditional neural network activations utilize fixed shape nonlinearities that apply global transformations to input distributions. We propose SentinelPulse-LAR (Localized Activation Residual), an activation mechanism that treats nonlinearity as a learnable, localized residual modification of the identity mapping. By decomposing the activation into an identity path and a Gaussian enveloped residual, the network can learn specific local corrections governed by a per activation amplitude α and a stage shared locality parameter β . Experiments on CIFAR-100 demonstrate that learnability of the residual amplitude plays a central role in performance gains. Beyond empirical results, this work is primarily mechanism focused.

Index Terms—Activation Functions, Neural Architecture, Residual Learning, Local Credit Assignment, SpLR

I. INTRODUCTION

The standard paradigm of activation functions such as ReLU [1], GELU [2], and Mish [3] relies on static mathematical curves. While effective, these functions apply fixed global nonlinearities that are agnostic to layer specific representational needs, placing the burden of adaptation primarily on weight matrices.

We introduce SentinelPulse-LAR (SpLR), which re-frames the activation as a micro residual operation. Instead of replacing the input signal x , the mechanism preserves identity by default and applies a signed, localized correction only where beneficial. This design follows a principle of minimal functional intervention, allowing nonlinearity to emerge through learnable, bounded deviations rather than global transformations.

II. MECHANISM IDENTITY AND CLAIMS

We define SentinelPulse-LAR as a mechanism-focused contribution based on the following claims:

- **Claim 1:** Identity preservation in the asymptotic regime ($|x| \rightarrow \infty$) ensures that large activations are passed through unchanged.
- **Claim 2:** The nonlinearity is explicitly localized through a learnable locality parameter β .
- **Claim 3:** Learnability of the residual amplitude α is required to align localized curvature corrections with layer specific representations during training.
- **Claim 4:** The mechanism is structurally distinct and non redundant compared to layer level residual blocks.

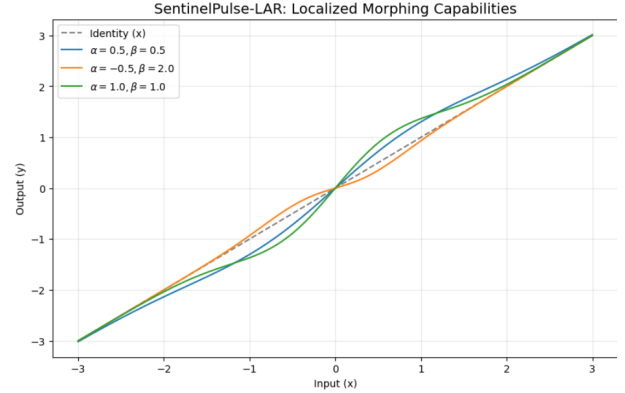


Fig. 1. Local morphing behavior of SentinelPulse-LAR. The activation preserves identity in the asymptotic regime while modulating curvature near the origin through learnable parameters α and β .

III. SENTINELPULSE-LAR MECHANISM

A. Mathematical Definition

SentinelPulse-LAR (SpLR) is defined as a localized perturbation of the identity mapping:

$$y = x + \alpha \cdot x \cdot \exp(-\beta x^2) \quad (1)$$

The function decomposes into a primary identity path and a residual correction $r(x) = \alpha x e^{-\beta x^2}$. In the asymptotic regime ($|x| \rightarrow \infty$), the residual term vanishes, and the mapping approaches the identity.

Fig. 1 illustrates how SentinelPulse-LAR introduces localized curvature corrections while preserving identity in the tails.

B. Parameterization and Golden Initialization

To ensure numerical stability, we adopt the constrained parameterization:

$$\alpha = 2 \tanh(\alpha_{\text{raw}}), \quad \beta = 0.01 + \text{softplus}(\beta_{\text{raw}}) \quad (2)$$

The canonical *Golden Initialization* is defined as $\alpha_{\text{raw}} = 0.182$ and $\beta_{\text{raw}} = -1.35$, yielding a mild initial correction that can be refined during training.

C. Gradient Behavior

The derivative of SpLR with respect to x is:

$$\frac{dy}{dx} = 1 + \alpha e^{-\beta x^2} (1 - 2\beta x^2) \quad (3)$$

This formulation preserves identity aligned gradient flow in the asymptotic regime while enabling localized curvature modulation near the origin.

IV. EXPERIMENTAL RESULTS

A. Vision Domain: CIFAR-100

We evaluated SpLR against Mish using identical architectures and training protocols across three random seeds.

TABLE I

CIFAR-100 PERFORMANCE (MEAN $\pm$ STD OVER 3 RANDOM SEEDS). ALL MODELS ARE TRAINED FROM SCRATCH USING THE SAME CONVOLUTIONAL NEURAL NETWORK ARCHITECTURE, DIFFERING ONLY IN THE ACTIVATION FUNCTION.

Activation	Epochs	Dropout	Val Acc (%)	Test Acc (%)
Mish	25	0.10	53.25 $\pm$ 0.21	56.17 $\pm$ 0.09
SpLR	25	0.10	54.50 $\pm$ 0.38	58.49 $\pm$ 0.21
Mish	50	0.10	56.26 $\pm$ 0.32	57.90 $\pm$ 0.30
SpLR	50	0.10	59.02 $\pm$ 0.22	61.75 $\pm$ 0.32

B. Ablation of Learnability

To isolate the effect of residual amplitude learnability, we evaluate reduced capacity diagnostic configurations rather than the full CIFAR-100 training setup used in Table I.

This design choice is also supported by the mathematical structure of the activation. SentinelPulse-LAR is defined as

$$y = x + \alpha x e^{-\beta x^2}, \quad (4)$$

with derivative

$$\frac{dy}{dx} = 1 + \alpha e^{-\beta x^2} (1 - 2\beta x^2). \quad (5)$$

The residual amplitude α directly scales both the forward residual magnitude and the deviation of the gradient from identity. When α is frozen, the curvature profile of the activation is fixed while the distribution of pre-activation values evolves during training, leading to a persistent mismatch that must be compensated by weight updates.

Allowing α to be learned restores alignment between curvature strength and evolving signal statistics. However, assigning independent α parameters to each activation introduces uneven scaling of the residual correction across neurons within a layer, increasing sensitivity to initialization and optimization noise. In contrast, stage-shared amplitude learning preserves coherent curvature modulation across the layer while retaining adaptivity. This mathematical structure explains the observed stability of the stage shared formulation and motivates its selection as the canonical configuration.

V. DISCUSSION

SentinelPulse-LAR introduces nonlinearity through a residual correction term

$$r(x) = \alpha x e^{-\beta x^2}, \quad (6)$$

which alters the activation behavior near the origin while leaving large inputs unchanged. The effectiveness of this mechanism depends critically on the ability of the residual amplitude α to co-adapt with evolving layer representations.

When α is learnable, the activation function participates directly in representation learning by jointly optimizing both the network weights and the strength of the residual correction. This can be seen by examining the input gradient of the activation:

$$\frac{dy}{dx} = 1 + \alpha e^{-\beta x^2} (1 - 2\beta x^2). \quad (7)$$

Here, α explicitly scales the deviation from identity in both the forward signal and its gradient. As training progresses, updates to α allow the activation to reshape local curvature in response to layer specific signal statistics.

In contrast, freezing α fixes the magnitude of the residual term and its gradient contribution. As a result, the activation introduces a static, localized deformation that cannot adjust to changes in the distribution of pre activation values. Weight updates must then globally compensate for this fixed local mismatch, leading to inefficient credit assignment. This explains the sharp performance degradation observed in the frozen amplitude variant.

SentinelPulse-LAR thus operates not as a static nonlinearity, but as a learnable micro residual module embedded at the activation level.

VI. CONCLUSION AND FUTURE WORK

We introduced SentinelPulse-LAR (SpLR), an identity preserving activation mechanism that treats nonlinearity as a localized, learnable residual. Experimental results demonstrate from scratch viability and highlight the importance of residual amplitude learnability. Future work will explore adaptive locality schemes and activation level optimization strategies that may reduce reliance on global backpropagation.

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